ELEC5620 Project Description

Background

With the rapid development of large language models, intelligent agents are becoming capable of planning, decision-making, tool use, and collaboration. This makes it possible for one person to operate an AI-powered virtual company supported by multiple specialised agents.

The theme of this project is **“One-Person AI Company.”** Students will design and implement an LLM-powered multi-agent system in which agents take on different business roles, collaborate to complete tasks, use external tools, and adapt their actions based on results. Through problem definition, system modelling, implementation, testing, and evaluation, students will gain practical experience in applying model-based software engineering techniques to intelligent agent systems.

Note: The related software applications include: **1) Healthcare 2) Finance; 3) Smart Home; 4) Retail and E-commerce; 5) Smart Personal Assistant; 6) Smart Cities and 7) Education.** You can choose any of the above applications.

Submissions

Stage 1: Requirements Documentation, Architecture Design and System Modeling (30 Marks)

Document the whole requirements, features, design and modelling process. A document should be submitted via Canvas, which should include:

1. (15 Marks) A formal report document that contains:

- A clear classification of the agreement, mandatory capabilities and optional features.
- User requirement diagram and Feature diagram.
- A description of the use cases for the principal features, including steps detailing how users and processes interact with the system.
- A description of the proposed system structure and its life-cycle processes; that is, the set of high-level components, their interconnections and interdependencies (package diagrams), class diagrams, collaborations, structured class, etc.
- A description of the behaviours of the key components, including activity diagrams, state machines and sequence diagrams, with a particular focus on the role of LLMs in the agent system. A brief description of the support components should be provided.
- Relationships between the key components should be provided to show support for future development work.

- The assumptions made during the design.

2. (15 Marks) Presentation and interview. Presentation slides should cover the following:

    - The proposed requirements and their classifications (to facilitate the finalisation of the requirements).
    - The proposed design, including the rationale for the design choices made (including discarded choices).
    - Overview of the models.
    - The basis for change assessment and acceptance determination.

3. The contribution of each group member should be specified clearly in the document.

Stage 2: Prototype Implementation and Demonstration (20 Marks)

In this stage, a proof-of-concept prototype should be implemented based on the architecture design. The prototype will be demonstrated to the stakeholders (tutors). For every group, members should work together to practice the Agile software development model, which will be discussed in the lectures. The prototype should:

1. Develop based on the requirements and design in Stage 1 and implement most models.
2. Demonstrate how the LLM enhances the agent's perception, decision-making, and interaction capabilities within the system.
3. Reflect the experience of using the Agile software development model.
4. Utilize advanced technologies in the software.

Note: Demonstration will be held in the final week. The project source code should be uploaded to Canvas before Monday of the final week.

Good luck!

ELEC5620 Teaching Team